

Bindu

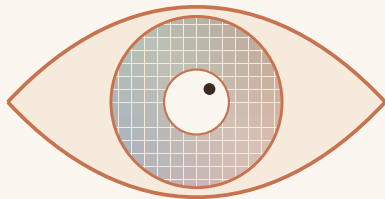
An image workspace built on
two-dimensional signals

Sajiid Jawad Talukder

2305041

Proteek Rosul

2305042



$x[m,n]$ $X[k,l]$

A picture is just a signal

- Every photo is a grid of samples $x[m, n]$, but photo apps **hide the maths** behind a slider.
- We wanted one workspace where each button is a **signal operation** you can see, measure and undo.
- It is built from the course toolbox: **convolution, the 2D DFT, linear transforms, wavelets** and entropy coding.

QUESTION → TOOL

How are colours spread?	histograms, luminance
Which frequencies?	2D FFT spectra
Blur or sharpen?	convolution kernels
Store it smaller?	Fourier pruning, Haar
Fix an old photo?	morphology, inpainting



Architecture: maths kept apart from the interface

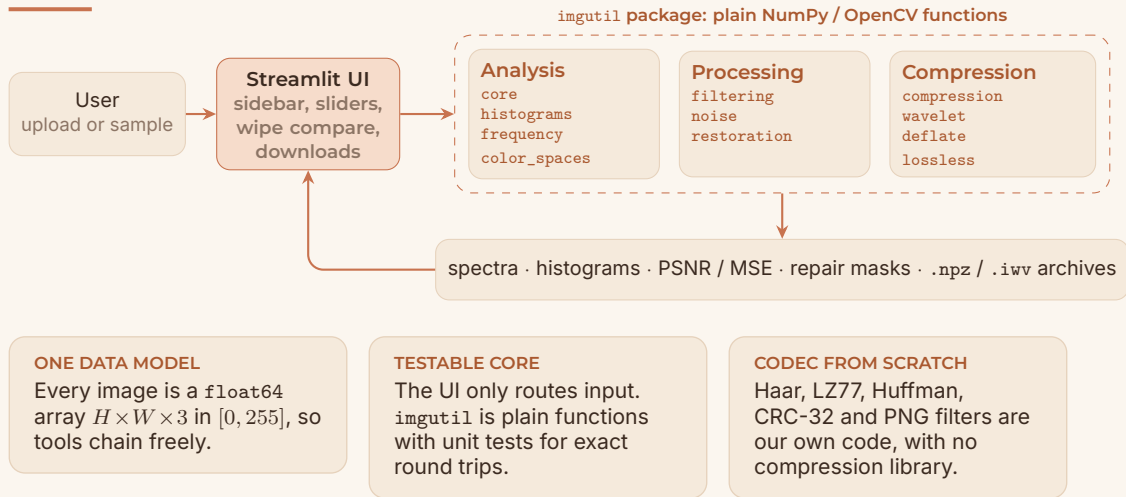
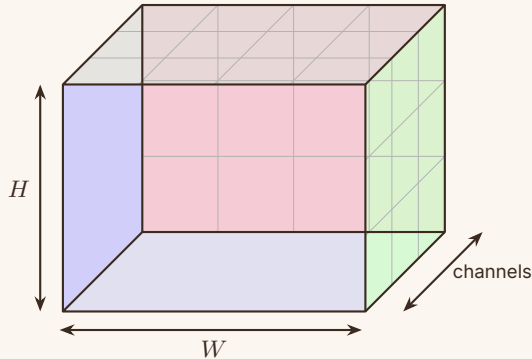
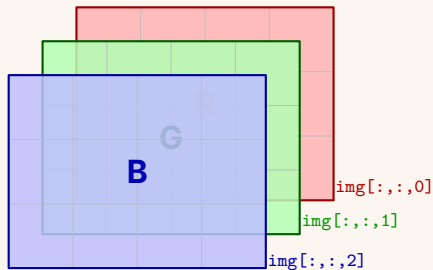


Image as a 3D array

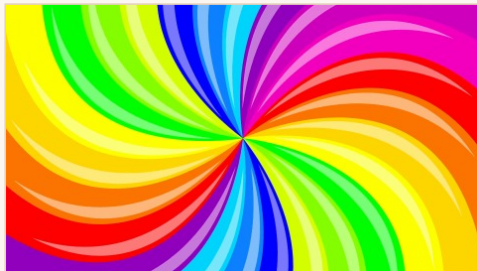
- A colour image is a 3D array of shape $H \times W \times 3$.
- The third axis indexes the three colour channels.



An image is three 2D signals stacked together



$$\begin{aligned} x_c[m, n], \quad c \in \{R, G, B\} \\ 0 \leq m < H, \quad 0 \leq n < W \\ x_c[m, n] \in [0, 255] \end{aligned}$$



Indexing the third axis, `img[:, :, c]`, splits the photo into its R, G and B planes.

Histograms: the statistics of each channel

HISTOGRAM OF CHANNEL c

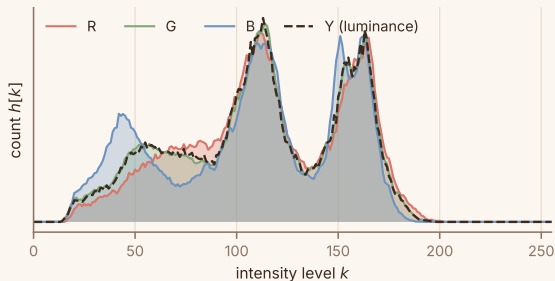
$$h_c[k] = \#\{(m, n) : x_c[m, n] = k\}$$

$$p_c[k] = \frac{h_c[k]}{HW}, \quad \sum_{k=0}^{255} p_c[k] = 1$$

LUMINANCE: ITU-R BT.601 FORMULA

$$Y = \begin{bmatrix} 0.299 & 0.587 & 0.114 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

Y uses the perceived luminance, which is closer to what the eye sees.



Lakeside sample. Blue sits highest (sky and water) and red lowest.

The dashed curve is Y : narrower, because the channels are averaged inside every pixel.

The 2D DFT: which frequencies build the picture

2D DISCRETE FOURIER TRANSFORM

$$X[k, l] = \sum_{m=0}^{H-1} \sum_{n=0}^{W-1} x[m, n] e^{-j2\pi(\frac{km}{H} + \frac{ln}{W})}$$

fftshift puts DC in the centre; we display $\log(1 + |X|)$.

PARSEVAL'S THEOREM

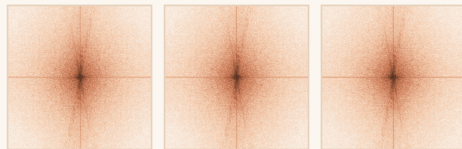
$$\sum_{m,n} |x[m, n]|^2 = \frac{1}{HW} \sum_{k,l} |X[k, l]|^2$$

Energy in space = Energy in frequency.

PARTIAL RECONSTRUCTION

$$\hat{x} = [\mathcal{F}^{-1}X_R, \mathcal{F}^{-1}X_G, 0]$$

LOG-MAGNITUDE SPECTRA, MEAN REMOVED



R

G

B

INVERSE FFT OF CHOSEN CHANNELS



all three

R+G

G+B

Energy crowds at the centre (low frequencies).

Colour space: RGB $\rightarrow$ YCbCr is a change of basis

FULL-RANGE BT.601 TRANSFORM

$$\begin{bmatrix} Y \\ C_b \\ C_r \end{bmatrix} = \begin{bmatrix} 0.299 & 0.587 & 0.114 \\ -0.169 & -0.331 & 0.500 \\ 0.500 & -0.419 & -0.081 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix} + \begin{bmatrix} 0 \\ 128 \\ 128 \end{bmatrix}$$

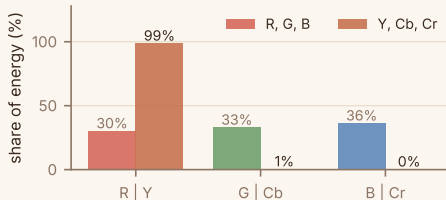
The C_b and C_r rows sum to zero, so a grey pixel ($R = G = B$) gets neutral chroma 128.

WHY IT HELPS

R, G and B move together ($\rho_{RG} = 0.994$ on this image). After the transform almost all the detail is in Y ; C_b and C_r are smooth, so there is less to store.



Y Cb (contrast $\times 5$) Cr (contrast $\times 5$)



Energy about the mean:
split evenly across R, G, B, but 98.9% goes to Y.

Filtering: convolution $\Leftrightarrow$ multiplication

2D CONVOLUTION = LTI SYSTEM

$$y[m, n] = \sum_i \sum_j h[i, j] x[m - i, n - j]$$

$$\Leftrightarrow Y[k, l] = H[k, l] \cdot X[k, l]$$

KERNELS ARE JUST DIFFERENT h

$$\frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

box blur: low-pass, $\sum h = 1$

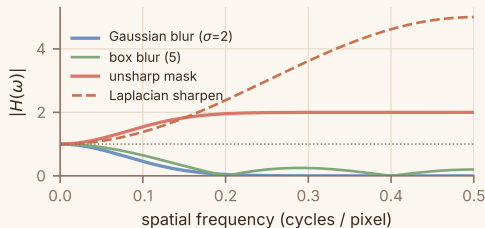
$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

sharpen: $x - \nabla^2 x$

GAUSSIAN AND UNSHARP MASK

$$G_\sigma \propto e^{-\frac{(i^2+j^2)}{2\sigma^2}}, \quad y = x + a(x - G_\sigma * x)$$

$$\Rightarrow \text{one kernel: } h = (1 + a)\delta - a G_\sigma$$



original

Gaussian, $\sigma=3$

unsharp, $a=1.5$

Linear, so per-channel = whole-image ($\max |\Delta| < 10^{-8}$).

Denoising: the right filter depends on the noise

TWO NOISE MODELS

Gaussian: $y = x + n$, $n \sim \mathcal{N}(0, \sigma^2)$

S&P: $y \in \{0, 255\}$ with prob. p , else x

MEDIAN FILTER (NON-LINEAR)

$y[m, n] = \text{median}_{|i|, |j| \leq 2} x[m + i, n + j]$

A lone 0 or 255 is never the middle value, so it is dropped.

NON-LOCAL MEANS

$$\hat{x}(p) = \frac{\sum_q w(p, q) x(q)}{\sum_q w(p, q)}$$

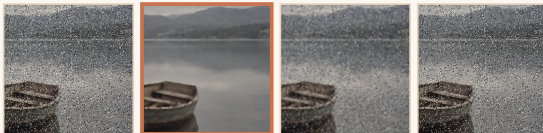
$$w(p, q) = e^{-\frac{\|P_p - P_q\|^2}{h^2}}$$

GAUSSIAN NOISE, $\sigma = 25$



noisy 20.2 dB median 30.0 Gaussian 30.3 **NL-means 31.2**

SALT & PEPPER, $p = 10\%$



noisy 15.8 dB **median 35.8** Gaussian 26.2 NL-means 16.2

PSNR vs. the clean crop (5×5 windows, $h = 15$).

Lossy: keep only the strongest frequencies

REAL IMAGE $\Rightarrow$ CONJUGATE SYMMETRY

$X[k, l] = X^*[-k, -l] \Rightarrow$ store only half.

ENERGY PRUNING, TARGET η

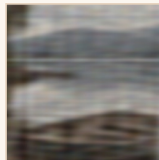
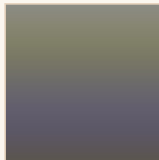
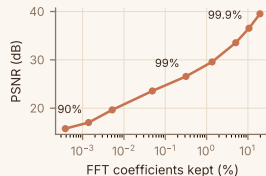
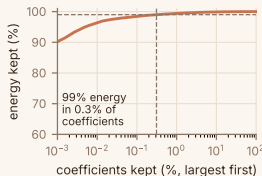
Rank pairs by $E_p = w_p |X_p|^2$; keep the fewest (DC always) with

$$\sum_{p \in S} E_p \geq \eta \sum_p E_p$$

$$\hat{x} = \mathcal{F}^{-1}\{X \cdot \mathbf{1}_S\}$$

MEASURING THE LOSS

$$\text{MSE} = \frac{1}{3HW} \sum (x - \hat{x})^2, \quad \text{PSNR} = 10 \log_{10} \frac{255^2}{\text{MSE}}$$



$\eta=90\% \cdot 17.0 \text{ dB}$

99% $\cdot 26.6 \text{ dB}$

99.9% $\cdot 36.5 \text{ dB}$

Our codec, 512 $\times$ 512 photo: 99% of the energy is in 0.32% of the coefficients. DC counts as energy, so detail returns near 99.9%.

Lossless compression I: integer Haar lifting

REVERSIBLE COLOUR TRANSFORM (INTEGERS ONLY)

$$\begin{bmatrix} R - G \\ G \\ B - G \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix}, \quad \det = 1$$

LIFTING ONE PAIR (a, b)

forward

$$d = b - a$$

$$s = a + \lfloor d/2 \rfloor$$

exact inverse

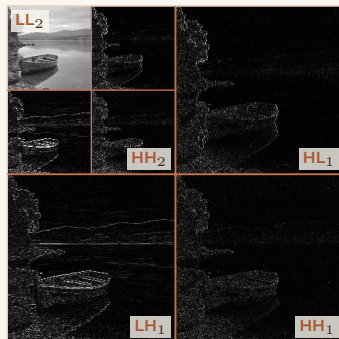
$$a = s - \lfloor d/2 \rfloor$$

$$b = a + d$$

EXAMPLE

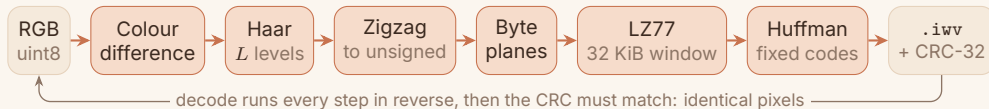
$$(10, 14) \xrightarrow{\text{lift}} (s, d) = (12, 4) \xrightarrow{\text{undo}} (10, 14)$$

Smooth areas give $d \approx 0$, and nothing is rounded away.



Two levels on our sample. Most detail is near zero (dark).

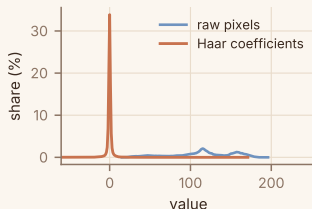
Lossless compression II: coefficients to a .iww file



ZIGZAG MAP

$$z = \begin{cases} 2c, & c \geq 0 \\ -2c - 1, & c < 0 \end{cases}$$

0, -1, 1, -2, 2 $\mapsto$ 0, 1, 2, 3, 4:
small magnitudes become small bytes.



ENTROPY, BITS PER VALUE

$$H = - \sum_v p(v) \log_2 p(v)$$

After the transform, 86% of values lie within ± 2 , so LZ77 and Huffman find far more repetition.

RESULT ON A 256×256 CROP

Entropy (pixels)	7.04 bits
Entropy (coefficients)	2.99 bits
Raw RGB	192.0 KiB
.iww archive	112.0 KiB
Ratio	1.71×
Decoded pixels	bit-exact

Restoration: find the scratches, then fill them

GREY-SCALE MORPHOLOGY, $b: 11 \times 11$ SQUARE

$$f \ominus b = \min_b f \quad \text{erosion}$$

$$f \oplus b = \max_b f \quad \text{dilation}$$

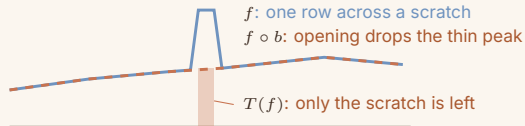
$$f \circ b = (f \ominus b) \oplus b \quad \text{opening}$$

$$T(f) = f - f \circ b \quad \text{white top-hat}$$

MASK, THEN REPAIR

$$M = [T(f) \geq \tau] \oplus (3 \times 3), \quad \tau = 30$$

Telea inpainting rebuilds the masked pixels from their neighbours. A mask over 10% of the image is refused.



damaged

top-hat T mask M

repaired

The mask covers 1.6% of the portrait. Bright hair highlights are picked up too, so a threshold slider controls the trade-off.

What Bindu delivers

DECORRELATION**98.9%**

of the variation
moves into Y after
one 3×3 matrix.

FOURIER PRUNING**0.32%**

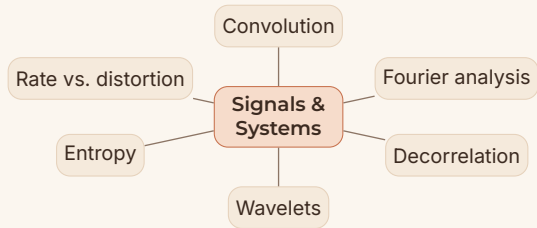
of the coefficients
hold 99% of the
energy (26.6 dB).

WAVELET CODEC**1.71 $\times$**

smaller than raw
RGB, and decoded
bit for bit.

RESTORATION**1.6%**

of the pixels
flagged and
inpainted on the
scratched portrait.



- **8 tools**: analysis, reconstruction, colour spaces, filters, noise lab, lossy and lossless codecs, scratch repair.
- Every result is **measured**: PSNR, MSE, file sizes, entropy and exact-match checks.
- Limits: our pure-Python codec is slower than zlib, and repair masks are heuristic.

Live demo

1 upload a photo

2 spectra & histograms

3 compress / decompress

4 repair scratches

Sajiid Jawad Talukder (2305041) · Proteek Rosul (2305042)